

Review Questions – Mobile Computing

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Q1. Define Mobile Computing and discuss the core technologies that enable it.

Mobile Computing enables users to access network services anywhere and at any time, breaking the geographical bond between the user and the network. Enabling technologies:

- Cellular Communication (e.g., 4G, 5G)
 - Wireless Local Area Networks – WLANs (e.g., Wi-Fi)
 - Satellite Services for global coverage
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Q2. Compare and contrast the four types of communication devices.

- **Fixed/Wired:** Desktop computers. Use case: Office workstations.
 - **Mobile/Wired:** Laptop + telephone line/modem. Use case: Hotel wired internet.
 - **Fixed/Wireless:** Sensors in hard-to-wire locations. Use case: Wireless sensors in a museum.
 - **Mobile/Wireless:** Roam between networks while connected. Use case: Smartphone moving between 5G and Wi-Fi.
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Q3. Describe the main characteristics of mobile computing.

- Independence from Fixed Positions – users can move freely.
- Wireless Connectivity – connection via wireless links.
- Portability – lightweight and handheld devices.
- Unrestricted Mobility – devices move through different network environments.

Portability provides the greatest advantage as it enables the "Mobile Office" concept – allowing real-time data access at a customer's site, providing a competitive edge.

Q4. How can mobile computing and ad-hoc networking be critical during emergencies?

- Ambulances transmit patients' vital signs to hospitals in real-time.
 - In natural disasters, wired networks are destroyed; wireless/ad-hoc may be the only available communication.
 - Vehicles spontaneously form networks to warn others of accidents or hazards.
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Q5. Identify three applications of mobile computing.

- **Vehicles:** Digital Audio Broadcasting (DAB) for news/weather and ad-hoc networks for emergency info. Benefit: real-time hazard alerts on the move.
 - **Business:** "Mobile office" – salesman accesses latest company data during presentations. Benefit: up-to-date info anywhere.
 - **Infotainment:** Location-based travel guides and ad-hoc gaming. Benefit: context-aware data based on exact location.
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Q6. How do limited battery and signal instability impact mobile app design?

- **Battery:** Limits CPU and memory → apps must be highly optimized.
 - **Signal:** Movement and obstacles cause frequent disconnections, high bit error rates, and variable delays → apps must be adaptive to poor connection states.
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Q7. Discuss the security constraints of mobile computing.

- Portability makes devices easy to lose or steal, risking stored data.
- Radio transmissions travel through the air and can be intercepted more easily than wired signals.

Mobile systems must employ efficient encryption, robust authentication protocols, and other security measures to protect the air interface.

Q8. Explain the Physical Layer and Data Link Layer in a wireless environment.

- **Physical Layer:** Converts bits into signals; handles frequency selection, carrier generation, signal detection in noise, and modulation.
- **Data Link Layer:** Ensures reliable point-to-point connectivity; handles MAC (managing who speaks when on a shared frequency), multiplexing, and error correction.

Medium Access is critical because multiple devices share the same frequency, so collisions must be managed.

Q9. Contrast the Network layer and Application layer in mobile computing.

- **Network Layer:** Routes packets; handles Device Location and Handover (switching base stations without dropping the call).
 - **Application Layer:** Handles service location, multimedia support, and ensures apps are adaptive to network quality changes.
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Q10. Define a MANET and describe its unique characteristics.

A MANET is an autonomous collection of mobile users communicating over bandwidth-constrained wireless links without fixed infrastructure. The topology changes rapidly as nodes move.

- No central hub.
 - Every node acts as both a host (sending its own data) and a router (forwarding data for others).
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Q11. Compare a traditional cellular network with a MANET.

- **Infrastructure Network:** Relies on fixed base stations; if the tower fails, the network fails.
- **MANET:** No fixed parts; self-configuring and peer-to-peer.

MANETs are preferred in: military operations, disaster recovery zones, and car-to-car communication for safety alerts.

Q12. Why are traditional IP addresses insufficient, and how does Mobile IP solve this?

Traditional IP addresses embed the host's network location. If a host moves, it must change its IP, terminating all ongoing sessions.

Mobile IP allows a node to change its network attachment while maintaining all communications with the same IP address, using a Care-of Address (CoA) to route datagrams regardless of location.

Q13. Distinguish between Mobility and Nomadicity.

- **Mobility:** Node changes network attachment while maintaining all existing communications with the same IP address.
 - **Nomadicity:** Node moves but must terminate all existing communications first; restarts with a new IP address at the new location.
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Q14. Identify the three primary components of Mobile IP.

- **Mobile Node (MN):** The mobile device accessing fixed network services while moving.
 - **Home Agent (HA):** Router at the MN's home network; maintains a registry of the node's current location.
 - **Foreign Agent (FA):** Router at the visited network; assists in delivering packets to the MN.
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Q15. Describe how a packet is delivered to a Mobile Node on a foreign network.

- MN discovers a Foreign Agent (FA) via agent discovery.

- MN registers its location with its Home Agent (HA) through the FA; FA's IP = Care-of Address (CoA).
- HA intercepts all packets addressed to the MN's home address.
- HA encapsulates packets and tunnels them to the FA.
- FA decapsulates packets and delivers them to the MN.

Outbound packets from the MN use standard IP routing.

Q16. How does the Agent Discovery protocol work?

An extension of the ICMP router discovery protocol; allows the MN to detect movement and find a Foreign Agent. Two message types:

- **Agent Advertisement:** Home/Foreign agents periodically multicast to advertise their presence.
 - **Agent Solicitation:** MN multicasts to request an advertisement without waiting.
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Q17. Discuss the Lifetime Period in Mobile IP registration.

- Every registration has a negotiated lifetime period.
 - MN must re-register before expiry, or the registration is deleted.
 - If lifetime = infinity, no re-registration is needed.
 - When MN returns home, it registers with lifetime = zero to delete its mobility binding.
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Q18. Explain the hierarchical architecture of a cellular network (BTS, BSC, MSC).

- **BTS (Base Transceiver Station):** Primary access point; covers a 120° sector; handles speech encoding, encryption, TDMA multiplexing, and modulation.
 - **BSC (Base Station Controller):** Controls multiple BTSs; manages radio channel allocation, frequency administration, power measurements, and handovers between its BTSs.
 - **MSC (Mobile Switching Center):** Network core; handles multiple BSCs; manages call setup, routing, switching, inter-BSC/MSC handovers, and interacts with HLR and VLR databases.
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Q19. Compare Hard Handoff and Soft Handoff.

- **Hard Handoff:** Break-before-make – connection to old cell is broken before new one is established. Used in GSM.
 - **Soft Handoff:** Make-before-break – device is connected to both old and new cells simultaneously for a smoother transition. Used in CDMA.
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Q20. Describe the different cell types and the purpose of frequency reuse.

- **Macrocell:** 10–50 km – rural areas.
- **Microcell:** 1–2 km – urban environments.
- **Picocell:** ~200 m – indoor (malls).
- **Femtocell:** ~10 m – home or office.

Frequency Reuse: using the same frequencies in different cells following a specific pattern, maximizing spectral efficiency and supporting more users without interference.

Q21. Discuss 1-persistent, non-persistent, and p-persistent CSMA.

- **1-persistent:** If idle → transmit immediately (prob = 1). If busy → keep listening until idle.
 - **Non-persistent:** If idle → transmit immediately. If busy → wait a random time before sensing again.
 - **P-persistent (slotted):** If idle → transmit with probability p, or defer to next slot with probability (1-p).
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Q22. Explain DAMA and why it is more efficient than Aloha.

- **Aloha efficiency:** Classical = 18%, Slotted = 36% – collisions occur during data transmission.

- **DAMA efficiency:** up to 80% – divides communication into a reservation period and a transmission period.
- Stations request slots during the short reservation period (collisions may occur here).
- Actual data transmission is collision-free because slots are explicitly assigned.

In Reservation Aloha (satellites), once a station reserves a slot, future slots are implicitly reserved, guaranteeing a data rate.